

EXPERIMENT 12

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Question 1:

Question:

Design a trigger such that whenever there is an insertion on the student table, the currently inserted or deleted row should be printed as it is in the output console window.

Answer:

```
CREATE OR REPLACE FUNCTION fn_student_audit()
```

```
RETURNS TRIGGER
```

```
LANGUAGE plpgsql
```

```
AS $$
```

```
BEGIN
```

```
    IF TG_OP = 'INSERT' THEN
```

```
        RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %',
```

```
            NEW.id, NEW.name, NEW.age, NEW.class;
```

```
    RETURN NEW;
```

```
    ELSIF TG_OP = 'DELETE' THEN
```

```
        RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %',
```

```
            OLD.id, OLD.name, OLD.age, OLD.class;
```

```
    RETURN OLD;
```

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END IF;

RETURN NULL;

END;

\$\$;

Question 2:

Question:

Design a PostgreSQL trigger such that:

- Whenever a new employee is inserted into tbl_employee, a record should be added to tbl_employee_audit with the message:
"Employee name <emp_name> has been added at <current_time>"
- Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit with the message:
"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

Answer:

```
CREATE TABLE tbl_employee (  
    emp_id SERIAL PRIMARY KEY,  
    emp_name VARCHAR(100) NOT NULL,  
    emp_salary INT  
);
```

```
CREATE TABLE tbl_employee_audit (  
    sno SERIAL PRIMARY KEY,  
    message TEXT  
);
```

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CREATE OR REPLACE FUNCTION audit_employee_changes()

RETURNS TRIGGER

LANGUAGE plpgsql

AS \$\$

BEGIN

IF TG_OP = 'INSERT' THEN

INSERT INTO tbl_employee_audit(message)

VALUES (

'Employee name ' || NEW.emp_name ||

' has been added at ' || NOW()

);

RETURN NEW;

ELSIF TG_OP = 'DELETE' THEN

INSERT INTO tbl_employee_audit(message)

VALUES (

'Employee name ' || OLD.emp_name ||

' has been deleted at ' || NOW()

);

RETURN OLD;

END IF;

RETURN NULL;

END;

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```
CREATE TRIGGER trg_employee_audit  
AFTER INSERT OR DELETE  
ON tbl_employee  
FOR EACH ROW  
EXECUTE FUNCTION audit_employee_changes();
```

Testing:

-- Insert an employee

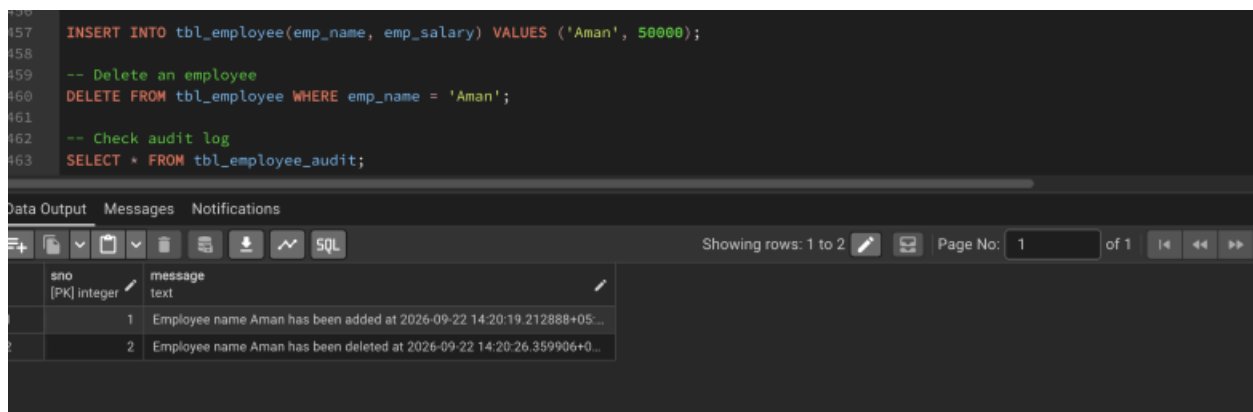
```
INSERT INTO tbl_employee(emp_name, emp_salary)  
VALUES ('Aman', 50000);
```

-- Delete an employee

```
DELETE FROM tbl_employee  
WHERE emp_name = 'Aman';
```

-- Check audit log

```
SELECT * FROM tbl_employee_audit;
```



```
157 INSERT INTO tbl_employee(emp_name, emp_salary) VALUES ('Aman', 50000);  
158  
159 -- Delete an employee  
160 DELETE FROM tbl_employee WHERE emp_name = 'Aman';  
161  
162 -- Check audit log  
163 SELECT * FROM tbl_employee_audit;
```

Data Output Messages Notifications

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sno	[PK] integer	message	text
1		Employee name Aman has been added at 2026-09-22 14:20:19.212888+05...	
2		Employee name Aman has been deleted at 2026-09-22 14:20:26.359906+05...	